COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE 010 | **SECTION:** | CPE21S4 |
| **DATE PERFORMED:** | July 8, 2026 | **DATE SUBMITTED:** | July 10, 2026 |
| **NAME:** | DE VERA, Sophia D. | **INSTRUCTOR:** | Engr. QUEJADO, Jimlord |

## ACTIVITY NO. 1.1 Algorithm, Flowchart, Pseudocode

### 1. PROCEDURE OUTPUT

- In this activity we took some review skills on how much we remember the basics of C++ Programming Language. But since we are running on a new environment/compiler, we first try to navigate and familiarize ourselves with its interface and commands.
- Firstly, entering the syntax and the header file library (the **Lines 1 and 2**). Then indicating the function (**int main()**) together with the code to be executed .
- The **cout** command is used to display the message needed to be displayed. In this case, the "**Hello World!**" is the expected outcome
- To end the program, the **return 0;** is used.

seatwork#1.cpp > main()

```
1   #include <iostream>
2   using namespace std;
3
4   int main()
5   {
6       int spider = 0;
7       cout <<"Hello World!";
8
9       return 0;
10  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\TIPQC\Documents\DSA De Vera_4-7pm> cd "c:\Users\TIPQC\Documents\DSA De Vera_4-7pm\" ; if ($?) { g++ seatwork#1.cpp -o seatwork#1 } ; if ($?) { .\seatwork#1 }
Hello World!
```

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Compute the total and average of 3 quizzes then display the total and average.

### CODE

SW1_Task1.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{

    int total_Qz;
    float overall_score, student_score;
    float HighestPossib = 0, total_score = 0;

    cout << "How many quizzes to be encoded: ";
    cin >> total_Qz;

    for (int i = 1; i <= total_Qz; i++) {

        cout << "Quiz Number: " << i << endl;

        cout << "Total Score: ";
        cin >> overall_score;

        cout << "Student's Score: ";
        cin >> student_score;

        HighestPossib += overall_score;
    total_score += student_score;
    }

    float average = total_score / total_Qz;

    cout << "Final Quiz Computation Average: ";
    cout << "Number of Quizzes: " << total_Qz << endl;
    cout << "Total Possible Score: " << HighestPossib << endl;
    cout << "Student Total Score: " << total_score << endl;
    cout << "Average Score: " << average << endl;

    return 0;
}
```

### OUTPUT

```
PS C:\DSA> C:\DSA\SW1_Task1.exe
How many quizzes to be encoded: 5
Quiz Number: 1
Total Score: 10
Student's Score: 9
Quiz Number: 2
Total Score: 15
Student's Score: 5
Quiz Number: 3
Total Score: 15
Student's Score: 7
Quiz Number: 4
Total Score: 25
Student's Score: 17
Quiz Number: 5
Total Score: 25
Student's Score: 20
Final Quiz Computation Average: Number of Quizzes: 5
Total Possible Score: 90
Student Total Score: 58
Average Score: 11.6
```

### ALGORITHM

1. START.
2. Declaring the variables to be used in the code:
   - Number of Quizzes to be encoded
   - The score in the quiz
   - Total score
   - Average score
3. Setting the default “total score” to 0.
4. The user will be asked how many quizzes will be encoded.
5. Loop until it has reached the number of quizzes entered.
   - During this time, the program asks the user to enter the quizzes’ score/s
   - Add the quizzes score/s to the total score
6. Compute the average score: Average = Total Score / Number of Quizzes
7. Display the “total score”.
8. Show the “average score”.
9. END OF PROGRAM.

## PSEUDOCODE

```
START

DECLARE the Quizzes_Total as integer.
DECLARE score, total, average as real.

Total Score = 0

DISPLAY "How many quizzes to be encoded: "
INPUT total_Qz

FOR Counter = 1 TO Quizzes_Total

    DISPLAY "Quiz Number: "
    Counter
    DISPLAY "Total Score: "
    INPUT overall_score
    DISPLAY "Student's Score: "
    INPUT student_score

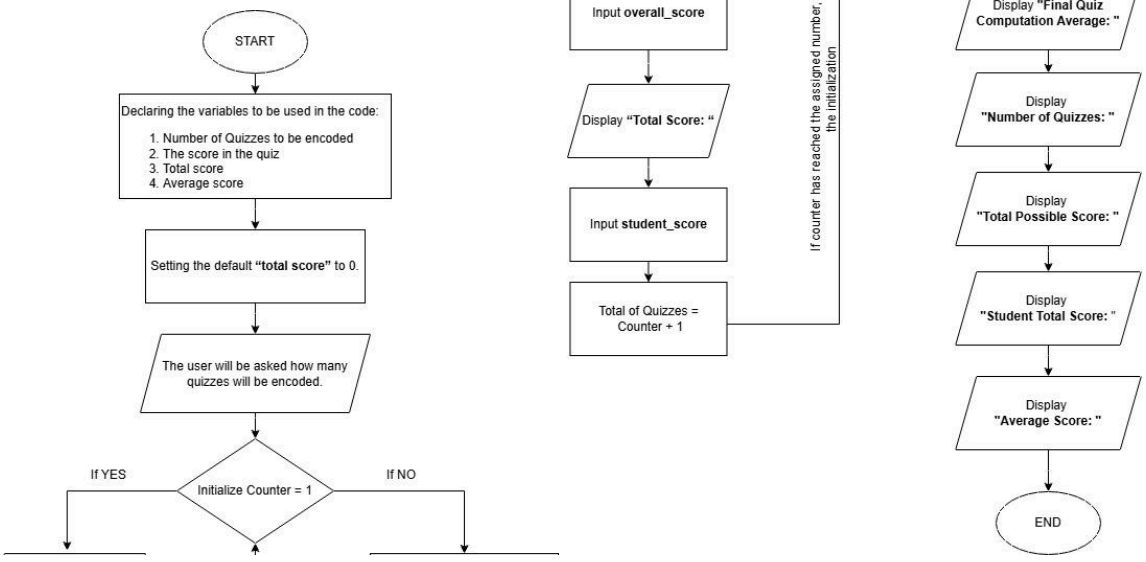
ENDFOR

average = total_score / total_Qz

DISPLAY "Final Quiz Computation Average: "
DISPLAY "Number of Quizzes: " , total_Qz
DISPLAY "Total Possible Score: " , HighestPossib
DISPLAY "Student Total Score: " , total_score
DISPLAY "Average Score: " , average

END
```

## FLOWCHART



2. Cashiering system that computes the total price of the 3 items bought, accepts payment and computes change. Display the transactions made.

## CODE

```cpp
int main()
{

    string groceries[6] = {
        "Pinoy Tasty", "Yakult", "Rebisco", "Soap", "Shampoo", "Sardines"};

    float prices[6] = {
        45.00, 48.50, 88.25, 72.20, 78.99, 23.00};

    string item;
    float total = 0, payment, change;
    bool found;

    cout << "\n";
    cout << "Billing Receipt"<< endl;
    cout << "Name of Product" << endl;
    cout << "Type 'total' when finished." << endl;
    cout << "\n";

    while (true) {
        cout << "Enter Product: ";
        getline(cin, item) ;

        if (item == "total")
            break;

        found = false;

         for (int i = 0; i < 6; i++)
         {
            if (item == groceries[i])
            {
                cout << groceries[i] << " PHP " << prices[i] << endl;
                total += prices[i];
                found = true;
                break;
            }
         }

         if (!found)
         {
            cout << "Product not found.\n";
         }
    }

    cout << "Total Amount: PHP " << total << endl;

    cout << "Payment: PHP ";
    cin >> payment;

    change = payment - total;

    cout << "Change: PHP " << change << endl;

    cout << "@DSA Grocery Shop.";

    return 0;
}
```

## OUTPUT

```
Billing Receipt
Name of Product
Type 'total' when finished.

Enter Product: Yakult
Yakult PHP 48.5
Enter Product: Yakult
Yakult PHP 48.5
Enter Product: Pinoy Tasty
Pinoy Tasty PHP 45
Enter Product: Shampoo
Shampoo PHP 78.99
Enter Product: Shampoo
Shampoo PHP 78.99
Enter Product: Rebisco
Rebisco PHP 88.25
Enter Product: Sardines
Sardines PHP 23
Enter Product: Sardines
Sardines PHP 23
Enter Product: soap
Product not found.
Enter Product: Soap
Soap PHP 72.2
Enter Product: total
Total Amount: PHP 506.43
Payment: PHP 1000
Change: PHP 493.57
@DSA Grocery Shop.
PS C:\DSA>
```

## ALGORITHM

1. START.
2. Declaring the variables used in the code:
   - Grocery Products
   - Products' Prices
   - Total amount
   - Customer's payment
   - Change (if there's any any)
   - Boolean Variable when a decision comes.
3. DISPLAY the billing receipt and instructions on what to encode.
4. Set the total amount to 0.
5. Loop until the user enters the keyword: total.
   - Customer enters the grocery product
   - Search the product name
   - If the product is not found:
     - Boolean Variable = FALSE
     - DISPLAY "Product not found."
   - If the product is found:
     - DISPLAY the grocery product's name with its equivalent price
     - Add the product to the total
     - Boolean Variable = TRUE
     - Then when the customer entered "total", exit the loop, and proceed to the next procedure
6. DISPLAY the total.
7. Ask the user for the payment amount.
8. Compute the change.
   - If the customer have change: Change = Payment – Total
9. END OF PROGRAM.

## PSEUDOCODE

```
START

DECLARE the Groceries[6] as a String
DECLARE the Prices[6] as a Real
DECLARE the Item as a String
DECLARE the Total, Payment, and Change as Real

Initializing Groceries and Prices
Total = 0

DISPLAY "Billing Receipt"
DISPLAY "Type 'total' when finished"

REPEAT

    DISPLAY "Enter Product: "
    INPUT Item

    IF Item = "total,
    THEN
        EXIT Loop
    ENDIF

    Found = FALSE
    Display "Product not found."

    ENDIF

    FOR Counter = 0 TO 5
    IF Item = Groceries[Counter]
    THEN
        DISPLAY  Groceries[Counter], Prices[Counter]
        Total = Total + Prices[Counter]
        Found = TRUE
        EXIT FOR
    ENDIF

ENDFOR

UNTIL Item = "total"
DISPLAY "Total Amount: "; Total

DISPLAY "Payment: "
INPUT Payment
```

```
Change = Payment - Total
DISPLAY "Change: "; Change

DISPLAY "@DSA Grocery Shop"

END
```

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

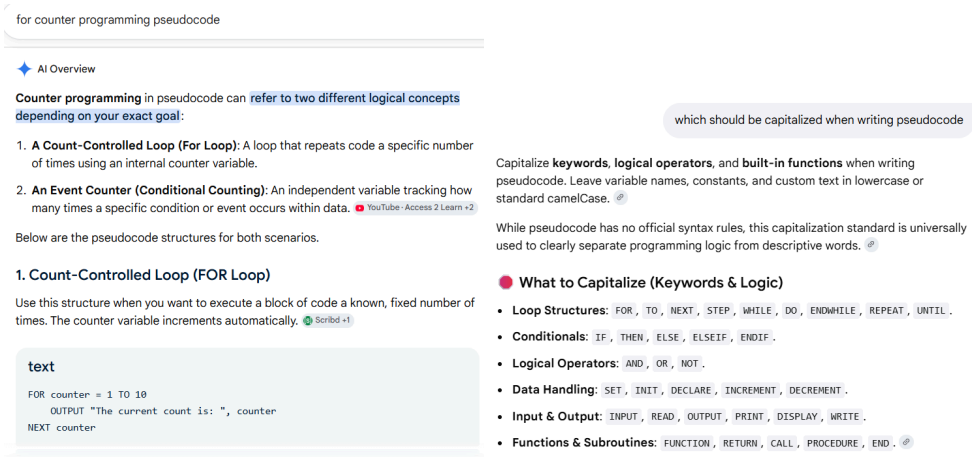
**Tools Used:** **Google AI Overview**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** **Asking for clarifications, explanation of concepts and as well as asking hints on what is wrong in my code.**

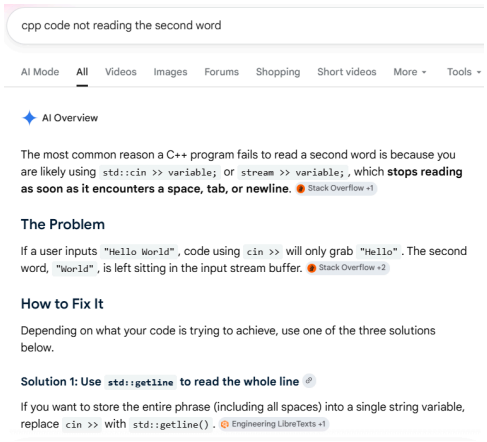
*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** **To make things clear and jog my memories on the topic I have difficulty remembering. *(here are the screenshots I have taken and recovered while doing the activity)***



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**REFERENCES:**


- **w3schools.com** https://www.w3schools.com/cpp/default.asp